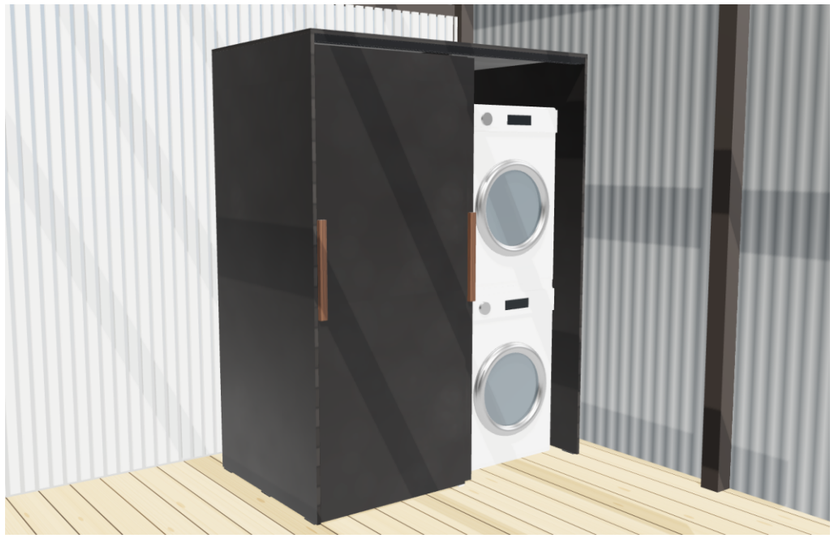
# Outdoor appliance cabinet: fridge + stacked washer/dryer





A plain black formply box: flush top, the fridge door flush with the sides, Cowdroy Quick-Fix door gear from Bunnings, no front vents, and two short American walnut pulls.

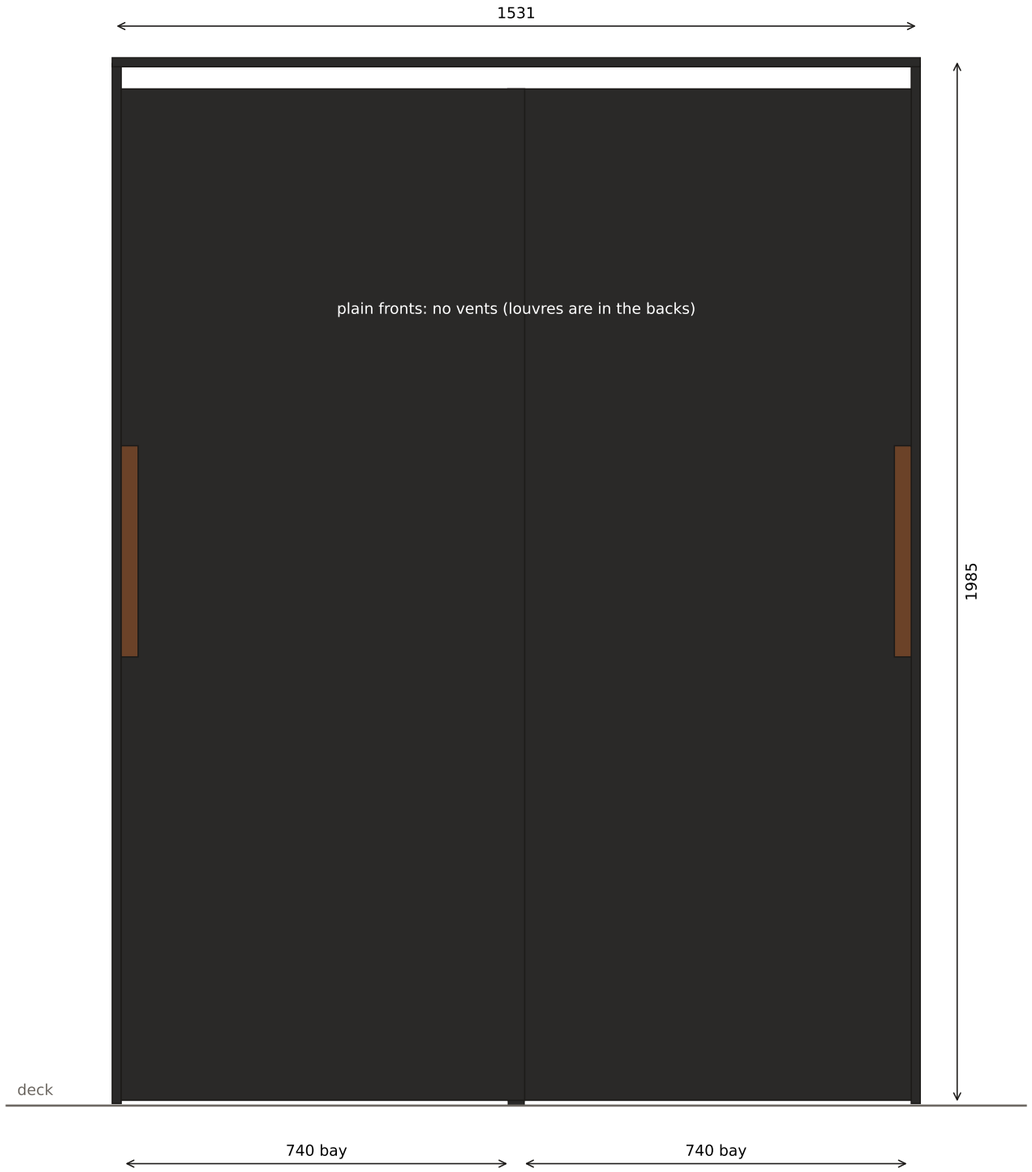
1531 W × 900 D × 1985 H. Two 740 clear bays. 8 CNC parts: 6 sheets of 17 mm formply, 2 of 19 mm for the doors.

The fridge breathes through flush louvred openings low and high in the hidden back panels (about 200 cm2 clear each level). Run the dryer with the laundry door slid open. The box sits on 4 mm ASA pads, held down by two stainless angles into the deck joists.

18 of 18 design checks pass (sheet 6). Open items must be closed before the cut.

# General arrangement

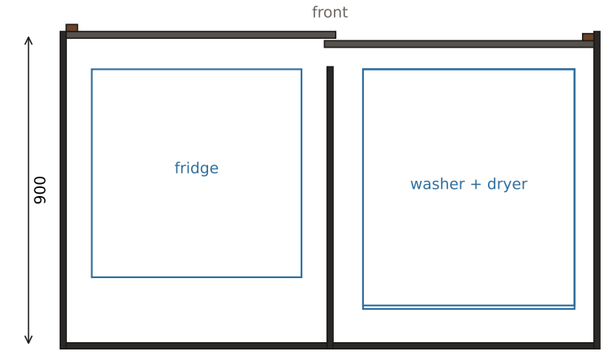
## Front elevation (doors closed)



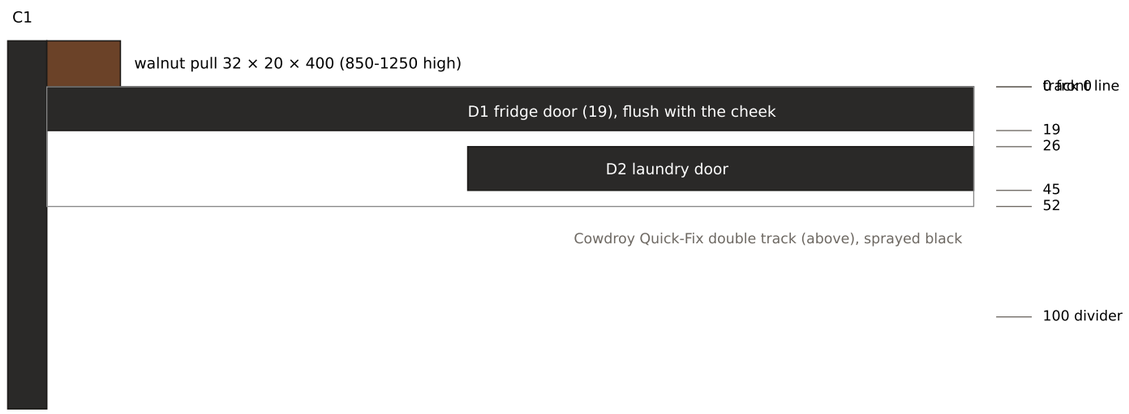

## Side elevation (studio cheek, outside)




## Plan (top removed)

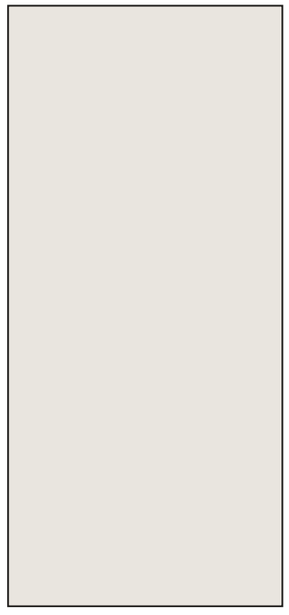



## Section through the doors and tracks (plan, enlarged)



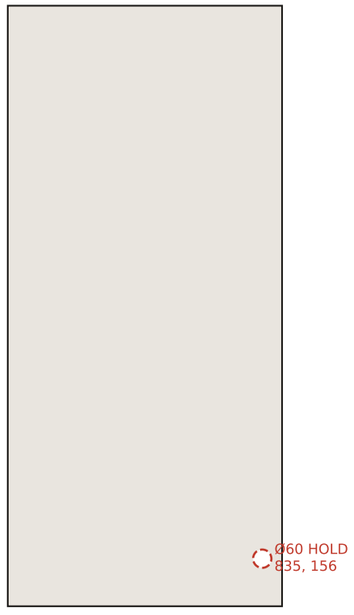
# CNC part drawings

C1 Left cheek  900 × 1970  ×1



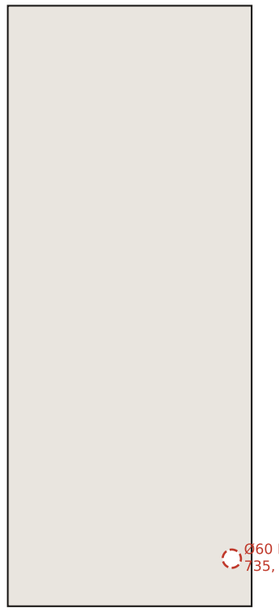
X = 0 at the cabinet front edge, Y = 0 at the bottom edge.

C3 Studio cheek  900 × 1970  ×1



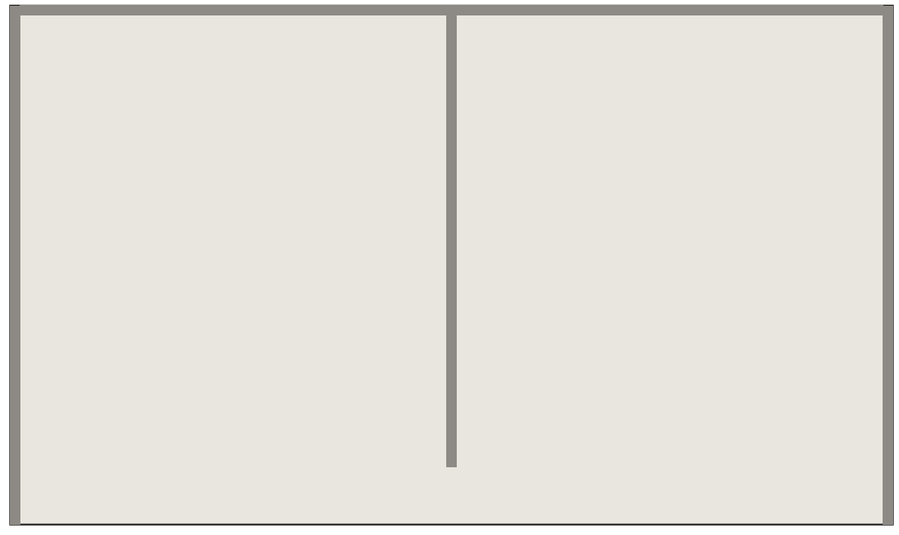

X = 0 at the cabinet front edge (mirror view: routed from the outside face). HOLD hole: confirm on site.

C2 Divider  800 × 1970  ×1



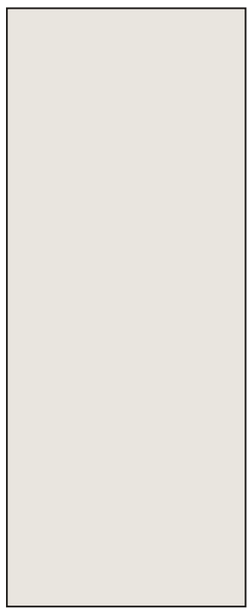

X = 0 at the front edge. HOLD hole: confirm on site.

C4 Top  1531 × 900  ×1



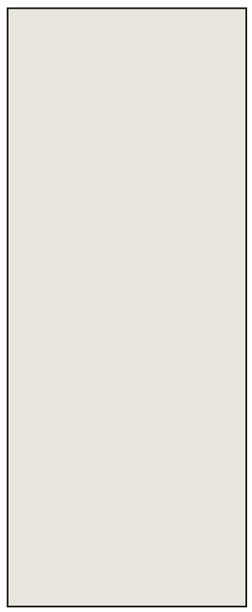
Routed on the UNDERSIDE: 6 mm housings for every vertical. X = 0 at the kitchen-side edge, Y = 0 at the front edge.

D1 Fridge door  764.5 × 1916  ×1



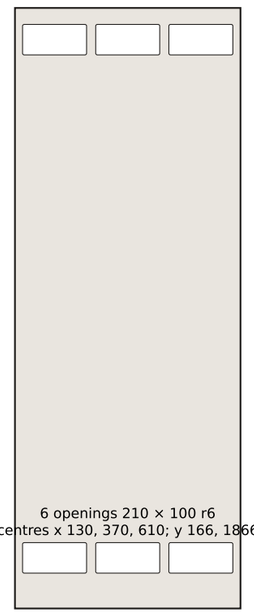
19 mm formply. Plain panel; the Quick-Fix wheel brackets screw to the top edge.

D2 Laundry door  764.5 × 1916  ×1



19 mm formply. Plain panel; the Quick-Fix wheel brackets screw to the top edge.

C5 Back, fridge bay  740 × 1970  ×1




6 openings 210 x 100, r6 corners, for the ASA louvre inserts. Drawn from the inside face, X = 0 at the left.

C6 Back, laundry bay: as C5.
C4 is routed on its UNDERSIDE: 6 deep housings (grey) for C1, C2, C3, the backs and the head rail.
Cheek housings are 0.6 over the panel for glue and fit.

# Cut list and indicative nests

| Code | Part | Size | Qty | Notes |
|---|---|---|---|---|
| C1 | Left cheek | 900 × 1970 × 17 | 1 | X = 0 at the cabinet front edge, Y = 0 at the bottom edge. |
| C2 | Divider | 800 × 1970 × 17 | 1 | X = 0 at the front edge. HOLD hole: confirm on site. |
| C3 | Studio cheek | 900 × 1970 × 17 | 1 | X = 0 at the cabinet front edge (mirror view: routed from the outside face). HOLD hole: confirm on site. |
| C4 | Top | 1531 × 900 × 17 | 1 | Routed on the UNDERSIDE: 6 mm housings for every vertical. X = 0 at the kitchen-side edge, Y = 0 at the [...] |
| C5 | Back, fridge bay | 740 × 1970 × 17 | 1 | 6 openings 210 x 100, r6 corners, for the ASA louvre inserts. Drawn from the inside face, X = 0 at the left. |
| C6 | Back, laundry bay | 740 × 1970 × 17 | 1 | As C5. |
| D1 | Fridge door | 764.5 × 1916 × 19 | 1 | 19 mm formply. Plain panel; the Quick-Fix wheel brackets screw to the top edge. |
| D2 | Laundry door | 764.5 × 1916 × 19 | 1 | 19 mm formply. Plain panel; the Quick-Fix wheel brackets screw to the top edge. |

S1  2400 × 1200 × 17



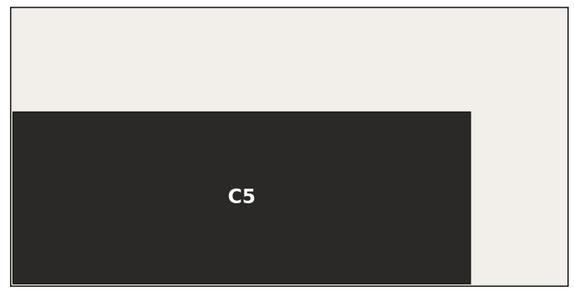

S2  2400 × 1200 × 17



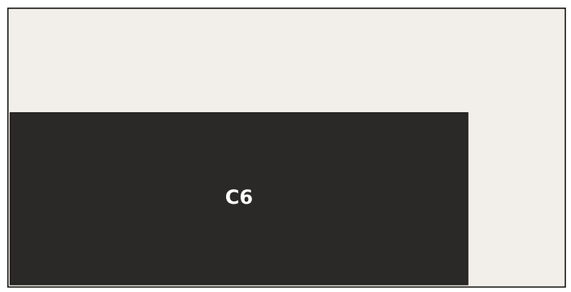

S3  2400 × 1200 × 17




S4  2400 × 1200 × 17



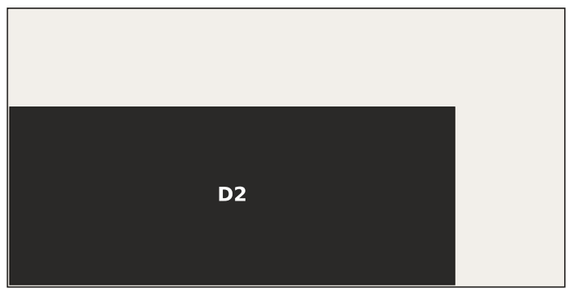

S5  2400 × 1200 × 17



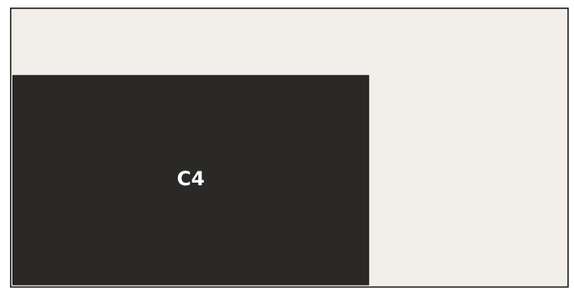

S6  2400 × 1200 × 19




S7  2400 × 1200 × 19




S8  2400 × 1200 × 17



# Build steps and shopping list

### 1. Confirm before cutting

Confirm the washer and dryer models and their stacking kit (dryer must be heat-pump or condenser). Buy the Cowdroy Quick-Fix set from Bunnings first and check its track height and the distance between the two door lines against the plans before the doors are released. Measure the power point position along the studio wall (it is about 100 above the deck) and the gaps to the slat screen and studio wall.

### 2. Seal and print

Seal every CNC-cut edge with two coats of exterior edge sealer, bottom and top edges of the doors most of all. Spray the Quick-Fix track satin black (etch primer first). Print the ASA parts (Prints tab).

### 3. Pads on the deck

Mark the footprint. Bed the thirteen 4 mm ASA edge pads in sealant on the lines of the cheeks, divider and backs, one 10g x 25 each.

### 4. Carcass on the top, upside down

Lay the top film-face down on padded trestles, housings up. Run PU sealant in each housing. Stand C1, C2, C3, C5 and C6 in their housings; the housings square the box. Screw down through the top into each edge, 8g x 50 stainless at 200 centres, countersunk and sealed.

### 5. Angles and louvres

Fit the four black aluminium angles inside the back corners, between the low and high vents. Push an ASA louvre insert into each of the twelve back openings from outside, blades sloping down and out, bedded in sealant, 2 x 6g x 16. Staple stainless mesh across the inside of each.

### 6. Stand it up and hold it down

Two people. Turn the carcass upright onto the pads. Check plumb and diagonals. Screw the two stainless hold-down angles inside the cheeks into the deck joists and the cheeks.

### 7. Track

Cut the Quick-Fix double J track to 1497 and screw it up into the top through its pre-drilled holes, front face in line with the cheek fronts so the fridge door hangs flush.

### 8. Doors and pulls

Screw the walnut pulls on from behind: on the left edge of D1 and the right edge of D2, 850-1250 above the deck. Screw two Quick-Fix wheel brackets to the top edge of each door, hang D2 on the rear line first and D1 on the front, and adjust the brackets to 10 off the deck. Fit a stop so D1 parks 40 short of the studio cheek.

### 9. Floor guide

Screw the Quick-Fix nylon floor guide to the deck where the doors cross (about 765 from the kitchen side). Run both doors end to end.

### 10. Grommets and leads

Fit the split grommets in the two low lead holes (160 above the deck) and run each original lead separately to the power point. No powerboard, nothing in a door track or under a machine foot.

### 11. Appliances in

Roll the fridge in, door hinge on the left, 107 back from the cabinet front. Stand the washer, fit the maker's stacking kit and lift the dryer on. Level all three on their own feet; the cabinet carries none of them.

| Type | Item | Qty | Use |
|---|---|---|---|
| Sheet | Black film-faced formply 2400 x 1200 x 17, exterior (A-bond) | 6 sheets | Carcass; CNC supplier supplies and cuts (Cut list tab) |
| Sheet | Black film-faced formply 2400 x 1200 x 19, exterior (A-bond) | 2 sheets | Doors (Quick-Fix needs 19 mm) |
| Sheet | Exterior edge sealer, black | 1 L | Two coats on every cut edge and the door grooves |
| Sheet | PU sealant/adhesive, black (Sikaflex-11FC or similar) | 4 x 300 ml | Housings, angles, louvres, pads |
| Hardware | Cowdroy 2400mm Quick-Fix Door Track Set (Bunnings I/N 3981776): double J track, wheel sets, nylon floor guide | 1 set, track cut to 1 | Top-hung bi-passing, 25 kg per door, 19 mm doors and up. Uses 4 of the 6 wheel sets |
| Hardware | Etch primer + satin black spray paint | 1 each | Track |
| Hardware | Aluminium angle 25 x 25 x 3, black anodised | 4 x 1700 | Inside the back corners |
| Hardware | Stainless angle 50 x 40 x 3 | 2 x 300 | Hold-downs inside the cheeks |
| Hardware | Stainless insect mesh | 1 m x 300 | Inside the twelve back openings |
| Timber | American walnut 32 x 20 x 400, from workshop offcuts | 2 | Door pulls |
| Timber | Exterior hardwood oil (e.g. Cutek or Intergrain UltraDeck Oil) | 250 ml | Walnut pulls; seal end grain |
| Fixings | 8g x 50 stainless countersunk timber screws | 60 | Through the top into the panel edges |
| Fixings | 8g x 25 stainless countersunk | 90 | Corner and hold-down angles |
| Fixings | 10g x 50 stainless countersunk | 4 | Hold-downs into the deck joists |
| Fixings | 10g x 25 stainless countersunk | 16 | Pads |
| Fixings | 6g x 16 stainless pan | 24 | Louvre inserts |
| Fixings | 8g x 30 stainless countersunk | 6 | Pulls, from behind |
| Fixings | Stainless staples, 10 mm | 1 box | Mesh |
| Print | ASA filament, black | about 2.5 kg | Prints tab: 3 STLs, 27 prints |

# Checks, open items and ASA prints

| | Check | Result |
|---|---|---|
| Pass | Fridge side clearance (maker 50) | 72.5 each side |
| Pass | Fridge back clearance (maker 50) | 186 to the back panel: the chimney behind the fridge |
| Pass | Fridge top clearance (maker 100) | 183 to the top; the tracks sit in front of the fridge face |
| Pass | Stack under the top | 238 above the dryer top |
| Pass | Laundry passes the open door | opening 841.5-1514, machines 859.0-1459.0 |
| Pass | Rear pull clear of the closed front door | 28 mm finger gap |
| Pass | Fridge door parks clear of the laundry pull | parked fridge door ends 8 short of the pull |
| Pass | Fridge opening with the door parked short | opening 17-709.5, fridge 89.5-684.5 |
| Pass | Rear pull in the gap between the doors | pull face 6 behind the front line |
| Pass | Quick-Fix limits (25 kg per door, 19 mm min) | 764.5 x 1916 x 19, about 18 kg |
| Pass | Track fits the opening | 1497 opening; Quick-Fix suits openings up to 1800 |
| Pass | Rear track clear of the divider | 48 mm |
| Pass | Leads exit near the power point | lead holes centred 160 above the deck; power point about 100 |
| Pass | Fridge vents (Liebherr/Electrolux built-in rule: 200 cm2 | per level per bay: 3 openings, 630 cm2 gross, about 207 net through louvre b |
| Pass | No straight path in for rain | louvre blades 45 degrees down and out, 27 deep at 13 pitch: each blade ove |
| Pass | High vents above the appliances | high openings start 1820; fridge 1785, stack 1730 |
| Pass | Doors clear of the deck | 10 above the deck |
| Pass | Fridge door opens 115 deg past the cheek | closest 24 mm at 115 deg (simplified slab, left hinge) |

| Open item | Before the cut |
|---|---|
| Washer and dryer models | Provisional 600 x 680 x 850 and 600 x 670 x 850. Confirm both models and the maker's stacking kit; the dryer must be heat-pump or condenser. |
| Dryer air | Run the dryer with the laundry door slid open. Makers say a cupboard dryer needs louvred door openings or the door open while it runs (LG asks for two door openings of 387 cm2). The back vents help but are not a substitute. |
| Quick-Fix geometry | The track height (30) and the gap between the two door lines (about 26) are taken from Cowdroy's published figures. Buy the set first and check both against the plans before the doors and the divider position are released. |
| Hardware outdoors | Quick-Fix is sold for indoor wardrobes. Under cover it should be fine; spray the wheels with dry PTFE lube once a year and keep the track clear of leaves. Replacement wheel sets are cheap and stocked. |
| Power point | About 100 above the deck on the studio wall; the leads now leave through 60 mm holes 160 above the deck. Measure its position along the wall. An outdoor outlet must be weatherproof (IP-rated). |
| Rear door set-back | With bypass doors one door has to run behind the other: the laundry door sits about 26 behind the flush fridge door. |
| Door flatness | 1916 tall formply doors rely on their balanced film faces to stay flat. Keep both faces sealed; if one bows, a black aluminium angle screwed up its back edge fixes it. |
| Washer door | With the laundry door open, the parked door sits just left of the washer and 60 to 80 back from its face. A left-hinged washer door opened past about 120 degrees will touch it. Prefer a right-hinged washer, or open it to 90 degrees. |

| ASA print | Qty | Size | Notes |
|---|---|---|---|
| 04_cable_grommet.stl | 2 | 82 × 82 × 20 | Split grommet for the two 60 mm lead holes (C2, C3). Smooth bore protects the leads. Print flange down. HOLD: confirm the plugs fit through 48 mm or enlarge the hole first. |
| 05_edge_pad.stl | 13 | 80 × 24 × 4 | 4 mm pad under the ply edges (13 of them). The box reads as flush to the deck but the ply never sits in water and the deck boards can dry. Bed in sealant, 1 x 10g x 25 into the deck. Print 100% infill. |
| 06_rear_louvre.stl | 12 | 230 × 120 × 30 | Flush louvre insert for each back opening (12). Only the 3 mm flange stands proud of the back. Blades slope 45 degrees down and out and overlap, so there is no straight path in for rain; water on a blade runs down and out. Push into the 210 x 100 opening from outside, bedded in sealant, 2 x 6g x 16. Staple stainless mesh across the inside. Print on its flange, no supports (45 degree blades). |